LUC@BJTU Weihai Campus

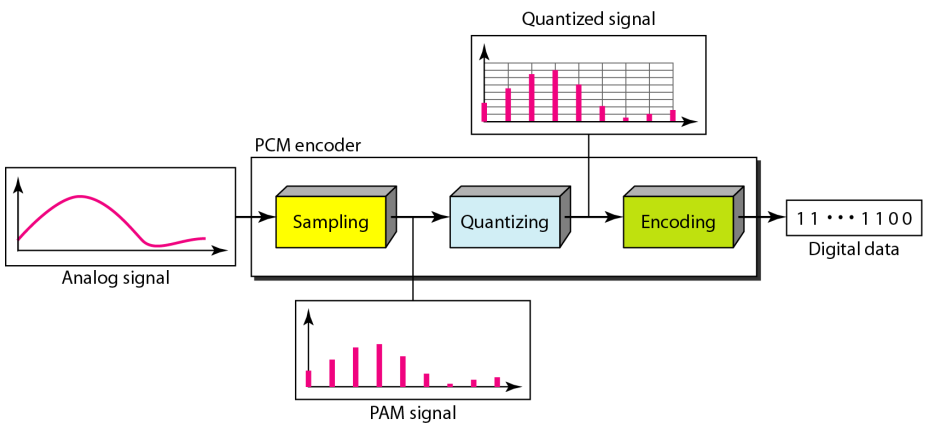
Fundamentals of Communication Systems (CNSCC160) Tutorial 6

1. Suppose we have an audio signal. How to convert it into a digital audio signal?

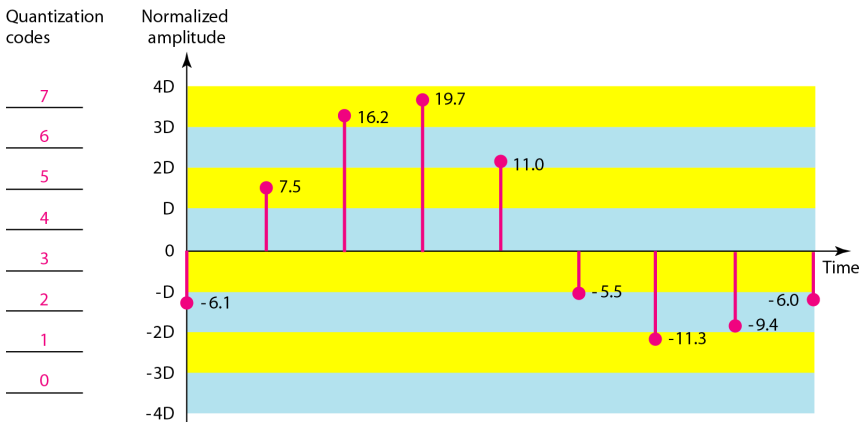
An analogue signal representing the sound/audio can be converted into digital audio signal with an analogue-to-digital converter (ADC), typically using pulse-code modulation (PCM).

This analogue-to-digital conversion process is more precisely called digitization. With PCM, the analogue signal is first sampled where the sampling rate need to satisfy Nyquist sampling theorem. Then, the sampled values will be quantized and converted to one of the n_b -bit codewords. Hence, PCM starts with a continuous-time, continuous-amplitude (analogue) signal, from which a digital signal is produced. The digital signal consists of blocks of n_b bits, where each n_b -bit number represents one amplitude of a PCM pulse.

PCM block diagram is given below.



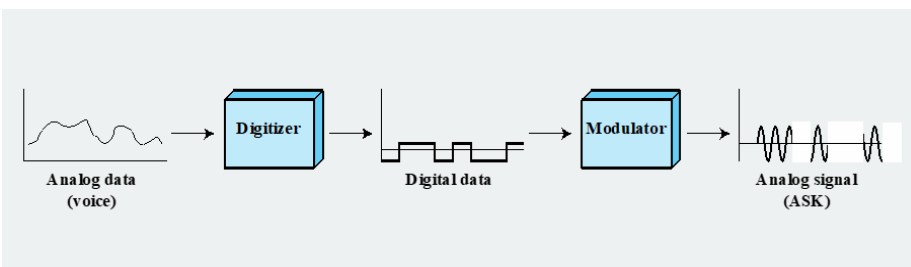
An example of digitization is shown below.



Normalized PAM values	-1.22	1.50	3.24	3.94	2.20	-1.10	-2.26	-1.88	-1.20
Normalized quantized values	-1.50	1.50	3.50	3.50	2.50	-1.50	-2.50	-1.50	-1.50
Normalized error	-0.38	0	+0.26	-0.44	+0.30	-0.40	-0.24	+0.38	-0.30
Quantization code	2	5	7	7	6	2	1	2	2
Encoded words	010	101	111	111	110	010	001	010	010

2. Once analogue audio is converted into digital audio, how to transmit it over the air?

The digital signal needs to go through the modulator to be modulated on a high-frequency carrier wave. The digital modulation techniques that can be used include ASK, FSK, PSK, etc.



3. We want to digitize the human voice containing frequency from 0 to 4000 Hz. What is the bit rate, assuming 8 bits per sample?

According to Nyquist sampling theorem, the sampling rate is 4000 * 2 which equals 8000 samples per second. Since bit rate equals the sampling rate times the number of bits per sample, we get that the bit rate = 8000 * 8 = 64 Kbps.

4. Why do we need multiple access protocols?

When nodes or stations are connected and use a common link, called a broadcast link, we need a multiple-access protocol to coordinate access to the link.

5. What are channelization (channel partitioning) protocols?

Channelization protocols (or channel partitioning, as it is sometimes called) are multiple-access methods in which the available bandwidth of a link is shared in time, frequency, or through code, among different stations/devices.

6. Explain the following terms: FDMA, TDMA, CDMA.

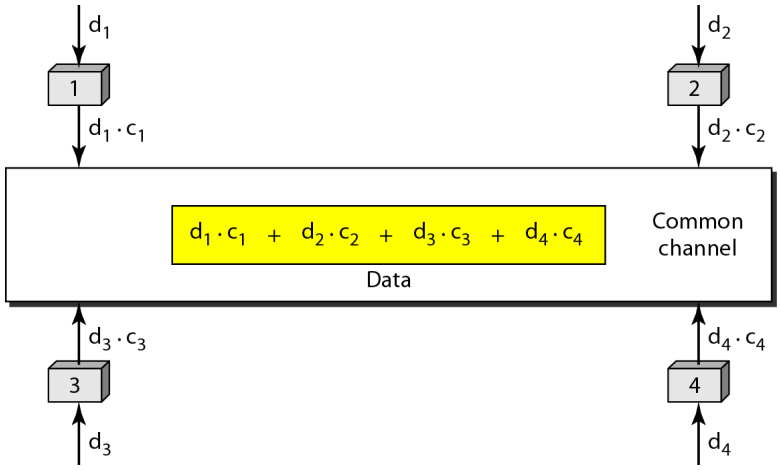
In frequency-division multiple access (FDMA), the available bandwidth is divided into frequency bands. Each station/device is allocated a band to send its data. In other words, each frequency band is reserved for a specific station, and it belongs to the station all the time.

In time-division multiple access (TDMA), the stations/devices share the bandwidth of the channel in time. Each station/device is allocated a time slot during which it can send data. Each station transmits its data in its assigned time slot.

In CDMA, a user has access to the whole bandwidth for the entire duration. The basic principle is that different CDMA codes are used to distinguish among the different users. CDMA codes need to be carefully chosen (some properties need to be satisfied).

7. Assume that there are four stations in total, connected to the same channel. The data from station 1 are d1, from station 2 are d2, and so on. The code assigned to the first station is c1, to the second is c2, and so on. Prove that a receiving station can get the data sent by a specific sender if it multiplies the entire data on the channel by the sender's chip code and then divides it by the number of stations. You can use the example given in the lecture to explain.

Let us prove this using the four-station example scenario introduced during our lecture.



We can say that the total data on the channel is D = (d1 · c1 + d2 · c2 + d3 · c3 + d4 · c4). The receiver which wants to get the data sent by station 1 multiplies these data by c1. Then, it gets that

$$
\begin{aligned}
D \cdot c_1 &= (d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4) \cdot c_1 \\
&= d_1 \cdot c_1 \cdot c_1 + d_2 \cdot c_2 \cdot c_1 + d_3 \cdot c_3 \cdot c_1 + d_4 \cdot c_4 \cdot c_1 \\
&= d_1 \times N + d_2 \times 0 + d_3 \times 0 + d_4 \times 0 \\
&= d_1 \times N
\end{aligned}
$$

When we divide the result by N, we get d1.

Note that each code is a sequence of numbers, for example, c1 = [+1 +1 +1 +1]. The above calculation is due to two main properties of chip codes:

1) The inner product of two equal sequences is N, i.e., c1 · c1 = N. For example, [+1 +1 +1 +1] · [+1 +1 +1 +1] = 1+1+1+1 = 4.
2) The inner product of two different sequences is 0, e.g., c1 · c2 = 0. Suppose that c2 = [+1 -1 +1 -1]. We can easily get that c1 · c2 = +1-1+1-1 = 0. This property is called orthogonality. Any two different codes are orthogonal.